Discussion Questions

1

Discuss the structure of the atom briefly, including

- an order of magnitude estimate of the size of the atom and its nucleus
- a comparison of the forces involved in the stability of the atom and the nucleus, including the range, strength and particles involved.

2

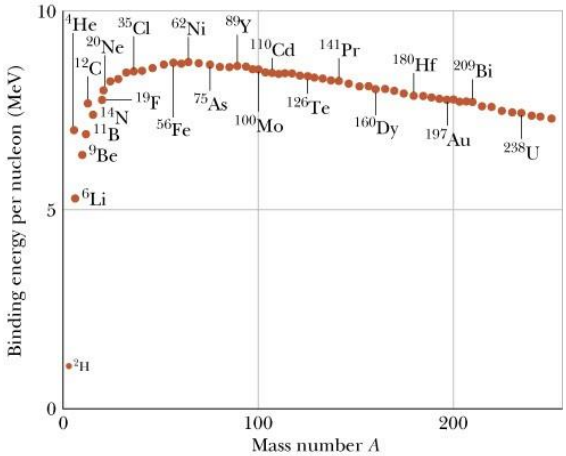
Over what range does the strong nuclear force play a significant role? How is this force important in explaining why the "line of stability" in elements is non-linear, i.e. why there are more neutrons in heavier elements than protons?

3

Explain the meaning of *binding energy.*

4

Explain the difference between fission and fusion, indicating the relevance of the graph below.



5

Wolfgang Pauli was led to suggest the existence of the *neutrino* long before its discovery, by a puzzle about the energy and momentum of the products of beta decay. What was this puzzle? Also, how did Fermi know that the neutrino should be a *neutral* particle?

6

In most observed galaxies, the orbital speed of stars around the galactic centre is constant over large radial distances, even far from the galactic core. Is this consistent with expected gravitational effects? What might this imply for the local mass density near the outer galactic rim? What solutions are proposed to explain any discrepancies?

Problem-solving Questions

7

The radioactive isotope ${}^{40}_{19}K$ undergoes beta (–) and beta (+) decay. Write the equation for both of these decays. Include the relevant neutrinos.

8

Another form of 'beta decay' is known as *electron capture*. In this process, an inner shell electron is 'captured' by the nucleus, with the resulting ejection of a neutrino from the nucleus. Find the isotope of the element labelled 'X' in the equation below:

$$ {}^{7}_{4}Be + {}^{0}_{-1}b \rightarrow X + n $$

9

A typical nuclear fission reaction of U-235 has the equation:

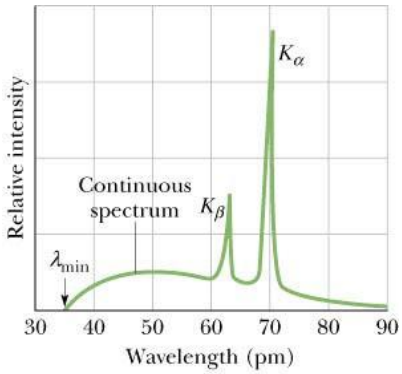
$$n + {}^{235}_{92}U \rightarrow {}_{56}Ba + {}^{89}_{36}Kr + 3n$$

• Find the isotope of Barium formed.
• the reaction above liberates ~ 175 MeV of energy. If 1 mole of U-235 fissions like this, estimate the amount of energy released, & compare it to the energy output of a 1 GW power station.

10

In an X-ray tube, high-energy electrons are fired at a target of *molybdenum* (Z = 42). The X-ray spectrum emitted is shown in the figure to the right. The continuous part of the spectrum is produced by the *bremsstrahlung* process (also called "braking radiation").

• Describe the bremsstrahlung process
• Calculate the energy of the shortest X-ray photon
• Explain the process which produces the two sharp peaks in the spectrum.
• Show that the energy of the K_α X-ray from an atom of element number Z is approximately equal to 10.2 $(Z-1)^2$ eV.



11

A hydrogen-like atom with $Z = 4$ and one electron is observed in an $n = 100$ state. Use Bohr theory to predict the size of the atom & the atom's energy (in eV).

[Bohr formulas for r and E: $r = \frac{a_B n^2}{Z}$, where $a_B = \frac{h^2 \varepsilon_0}{\pi m e^2}$; $E = -13.6 \frac{Z^2}{n^2} (eV)$]

Past examination questions

12

Consider a hydrogen atom initially in its ground state ($E = -13.6$ eV, $N = 1$), as shown in the figure to the right.

a) Copy the figure and label the energies of the four most tightly bound states.

A photon from an external source is absorbed and excites the atom to the 3rd excited state ($N = 4$).

b) Draw an arrow on the diagram that corresponds to this transition.
c) What is the energy of the photon in electron volts?
d) In which part of the electromagnetic spectrum (radio wave, microwave, infrared, ultra-violet, X-ray or gamma ray) was this photon?

From the $N = 4$ state, the atom decays by emitting a photon of wavelength 486nm.

e) Calculate the momentum of the emitted photon.

13

If the half-life for a radioactive nuclide is known, one can in principle use this to measure large time intervals. ^{40}K is one such element, decaying to ^{40}Ar with a half-life of 1.25×10^9 years. Analysis of moon rock samples reveals that the ratio of stable ^{40}Ar to the radioactive ^{40}K is 10:1.

a) Assuming that at formation of the rock there were N_0 atoms of ^{40}K, and zero atoms of ^{40}Ar, show that

$$\lambda t = log_e \left[1 + \left(\frac{N_{Ar}}{N_K} \right) \right]$$

where t is the time since the formation of the rock, and λ is the disintegration constant.

b) Use the relationship in (a) to determine the approximate age of the moon rock.

14

Listed below are *some* of the elements of the actinide series of the periodic table.

$$_{92}U \quad _{93}Np \quad _{94}Pu \quad _{95}Am \quad _{96}Cm \quad _{97}Bk$$

All of these are radioactive, and apart from uranium none are found naturally occurring in any significant qualities on Earth. In contrast, the $^{238}_{92}U$ isotope is found in abundance.

a) What can you infer about the half-life of $^{238}_{92}U$ in comparison with the age of the Earth?

Of the synthetically produced actinides, the isotope Americium ($^{241}_{95}Am$) is the only one to have made its way into our homes where it is commonly used in smoke detectors. Most $^{241}_{95}Am$ is produced as a by-product of the nuclear fission cycle by decay of the radioactive element $^{241}_{94}Pu$.

b) What type of decay produces $^{241}_{95}Am$ from $^{241}_{94}Pu$?

A smoke detector o contained 1.000 μg of americium when it was manufactured 30 years ago. The americium is also radioactive and decays by alpha particle emission with a half-life of 432.2 years.

c) Calculate the mass of $^{241}_{95}Am$ that remains in the smoke detector today.

d) Re-write the equation below by specifying the values of A and Z and identifying element X, to correctly reflect the radioactive decay of americium.

$$^{241}_{95}Am \rightarrow ^{A}_{Z}X + \alpha$$

15

The nuclear reactor at ANSTO creates the isotope Mo-99 which then decays to the metastable isotope Tc-99m. The Tc-99m is used as a radioactive tracer for helping to determine the cause of a patient's illness. The Tc-99m tracer emits a gamma ray during its decay to Tc-99. Tc-99 then further decays to Ru-99 which is a stable isotope.

The decay series and relevant information for each isotope is given in the table below;

Name		**Isotope**	**Mass (u)**	**Half-life**	**Decay**
Molybdenum	Mo-99	$^{99}_{42}Mo$	98.9077119	2.7474 days	Beta
Metastable Technetium	Tc-99m	$^{99m}_{43}Tc$	98.9064065	6.0058 hrs	Gamma
Technetium	Tc-99	$^{99}_{43}Tc$	98.9062547	211,000 yrs	Beta
Ruthenium	Ru-99	$^{99}_{44}Ru$	98.9059393	stable	

Rest masses of: $^{4}_{2}He$ atom: 4.002602 u; electron: 0.000549 u; positron: 0.000549 u

$^{1}_{1}H$ atom: 1.007825 u; neutron: 1.008665 u; proton: 1.007276 u

(a) The Molybdenum is transported to hospitals and produces the metastable Technetium to be injected into the patient.

(i) Mo-99 decays to the metastable Tc-99m via beta decay. Write the decay equation and justify whether the decay is β- or β+.

(ii) Show that the energy released during this decay is 1.216 MeV.

(b) Ruthenium is the only stable atom in the decay series. Determine the binding energy per nucleon of Ru-99.

Short Solutions

1 Size of atom (very rough) range 0.1 – 0.7 nm; nuclei size range: 2 fm – 15 fm; refer to Coulomb forces (long range); strong force (short range); refer to p, n and e.

2 Range$\sim 10^{-15}$ m. As SNF effects charged *and* neutral particles, while Coulomb force repulsion only effects charged particles, more neutrons help hold matter together.

3 *Binding energy* is the mass-energy difference between the mass-energy of the individual components of a nucleus and the mass-energy of the assembled nucleus; it is equal to the energy that would be required to 'unbind' the nucleus into individual protons and neutrons

4 Binding energy graph (on left) shows why fusion produces more energy than fission; increase in binding energy associated with decrease in mass, use $E = mc^2$.

5 Emitted products had a range of both energy and momentum; there was 'missing' energy and momentum; neutral because charges balanced

6 Unexpected – Gravity and Kepler predict slower distant orbits; implies more mass ; presence of Dark Matter or modification of Newtonian/Einsteinian gravitation theory.

7 ${}^{40}_{19}K \rightarrow {}^{40}_{20}Ca + {}^{0}_{-1}b + \bar{n}$ (beta –) and ${}^{40}_{19}K \rightarrow {}^{40}_{18}Ar + {}^{0}_{+1}b + n$ (beta +)

8 Li-7 (Z = 3)

9 Ba-144; about 17 TJ; output for 5 hours

10 scattering and acceleration of electrons from Mb atoms, range of decelerations, hence range of X-ray energies, max energy at shortest λ, use λ = 35 pm; sharp peaks are atomic transitions to 1s shell after inner shell electrons ejected; value comes from Mosely theory.

11 As a_B (the Bohr radius) = 5.29× 10–11 m; $r = 1.32 \times 10^{-7}$ m; $E = -0.0218$ eV

12 (**a**) **–**13.6 eV, –3.4 eV, –1.5eV, –0.85eV (**b**) arrow points upward from n= 1 to n = 4

12 (**c**) 12.75 eV (**d**) ultraviolet

13a Substitute $N_K = N_0 e^{-\lambda t}$ and $N_{AR} = N_0 - N_0 e^{-\lambda t}$ and do some algebra

13b 4.3×10^9 years (decay constant is 5.54×10^{-10})

14 (a) half-life is of the same order or longer than the lifetime of the Earth; if it was much shorter it would all have vanished (b) beta (minus) decay (should be obvious) (c) 0.953 μg (calculate λ for the decay from the half-life & then use $N = N_0 e^{-\lambda t}$) (d) ${}^{241}_{95}Am \longrightarrow {}^{237}_{93}Np + {}^{4}_{2}a$

15 (a)i. ${}^{99}_{42}Mo \rightarrow {}^{99m}_{43}Tc + {}^{0}_{-1}\beta + \bar{\nu}$ (a)ii. Consider number of electron-like particles before and after. Note Atomic masses *include* electron masses (b) 8.738 MeV per nucleon